Cosmic Censorship Violation at Large D

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Abstract: In this work, the Large-D limit of General Relativity is used as a framework to investigate violations of weak cosmic censorship and ultimately to reproduce the results of [3, 4]. In this limit, Einstein's equations simplify allowing gravitational dynamics to be described in terms of collective fields. A dimensional reduction is performed in order to work with spacetimes of an arbitrary number of dimensions. We study the stability of different objects in this limit and numerical simulations do show that, for sufficiently high angular momentum, there are no stable stationary black hole solutions in the theory. Finally, we provide evidence for a violation of weak cosmic censorship by simulating the collision of two boosted black holes.
Keywords: Gravitation, black holes, cosmic censorship hypothesis
SDGs: Quality Education

I. INTRODUCTION

General Relativity is the geometric theory of spacetime, extending special relativity to include gravitation as the manifestation of the curvature of spacetime.
The curvature of spacetime is described by Einstein's equations

$$R_{\mu\nu} - \frac{1}{2}g_{\mu\nu}R + \Lambda g_{\mu\nu} = 8\pi G T_{\mu\nu}. \tag{1}$$

These equations are highly nonlinear, consequently, only a limited number of analytical solutions are known. General Relativity has a natural parameter, the number of dimensions D, that we can vary and the theory remains well defined[1]. It is then natural that in order to deal with the complexity of Einstein's equations, we look for a limit of the parameter D in which the theory simplifies. The limit that will allow for a simplification of the theory is $D \to \infty$, the so-called large-D limit of General Relativity which will simplify enormously the equations[1].

General Relativity predicts the formation of spacetime singularities from dynamical settings such as gravitational collapse[2]. This led to the formulation of the Weak Cosmic Censorship (WCC) conjecture, which states that gravitational collapse cannot lead to the formation of a naked singularity[2]. This means that any spacetime singularity has to be hidden behind a black hole horizon, i.e., it lies in a causally disconnected region of spacetime. The large-D limit of General Relativity will allow us to study instances of violations of the WCC[3–5]. In this work a derivation of the large-D equations is presented and then these equations are used to explore WCC in black hole collisions.

II. DIMENSIONAL REDUCTION

In order to work with a spacetime of an arbitrarily large number of dimensions it is necessary to perform a dimensional reduction[6]. We therefore decompose the metric in its radial components and its spherical components

$$d\bar{s}^2_D = ds^2_{p+2} + e^{2\phi(x)} d\Omega_{n+1}. \tag{2}$$

We denote with a bar all D-dimensional quantities (e.g. $\bar{g}_{MN}$, $\bar{\Gamma}$, $\bar{R}$), while unbarred quantities refer to the reduced $(p+2)$-dimensional theory with $n \equiv D - p - 3$ and

$$ds^2_{p+2} = g_{\mu\nu}(x)\, dx^\mu dx^\nu, \qquad d\Omega_{n+1} = \gamma_{ab}(y)\, dy^a dy^b. \tag{3}$$

with $\gamma_{ab}(y)$ being the metric of the unit $(n+1)$-sphere, the brane indices run $\mu = 0, 1, \ldots, p+1$ and the sphere indices run $a = p+2, \ldots, D-1$. This decomposition allows us to construct a theory of gravity in $p+2$ dimensions coupled to a scalar field $\phi(x)$. We can now compute the components of the Riemann tensor(the detailed computations can be found in Appendix A)

$$\bar{R}_{\mu\nu\rho}{}^{\sigma} = R_{\mu\nu\rho}{}^{\sigma}, \tag{4}$$
$$\bar{R}_{abc}{}^{d} = R_{abc}{}^{d} + g^{\mu\rho} e^{2\phi} \nabla_\rho\phi \nabla_\mu\phi \left(\gamma_{bc}\delta^d_a - \gamma_{ac}\delta^d_b\right), \tag{5}$$
$$\bar{R}_{\mu a\nu}{}^{b} = -\delta^b_a \left(\nabla_\mu\nabla_\nu\phi + \nabla_\mu\phi\nabla_\nu\phi\right), \tag{6}$$
$$\bar{R}_{a\mu b}{}^{\nu} = g^{\nu\rho}\gamma_{ab} e^{2\phi}\left(\nabla_\mu\phi\nabla_\rho\phi + \nabla_\mu\nabla_\rho\phi\right), \tag{7}$$

with all other mixed components equal to zero.

Finally, the Ricci tensor is

$$\bar{R}_{\mu\nu} = R_{\mu\nu} - (n+1)\left(\nabla_\mu\nabla_\nu\phi + \nabla_\mu\phi\nabla_\nu\phi\right), \tag{8}$$
$$\bar{R}_{ab} = R_{ab} - g^{\mu\rho}\gamma_{ab} e^{2\phi}\left(\nabla_\mu\nabla_\rho\phi + (n+1)\nabla_\mu\phi\nabla_\rho\phi\right), \tag{9}$$

and the mixed Ricci components vanish.
The contribution of the $(n+1)$-sphere to the curvature of our spacetime is encoded in what we will call warp factor

$$W_{\mu\nu} = -(n+1)\left(\nabla_\mu\nabla_\nu\phi + \nabla_\mu\phi\nabla_\nu\phi\right). \tag{10}$$

III. EFFECTIVE EQUATIONS AND GREGORY LAFLAMME INSTABILITY

In this section we specialize to black strings, which correspond to a one-dimensional spatial worldvolume ($p = 1$). A brief remark on notation, to capture the relevant physics, physical quantities will be rescaled in the effective theory. We denote physical quantities using boldface, while keeping the same symbols.
In order to obtain a solution for an asymptotically flat neutral black string, it is useful to consider a Schwarzschild–Tangherlini solution and add one additional spatial coordinate σ[7]:

$$ds^2 = -f(r)\,dt^2 + \frac{dr^2}{f(r)} + d\boldsymbol{\sigma}^2 + r^2 d\Omega_{n+1}, \tag{11}$$

with $f(r) = 1 - \left(\frac{r_0}{r}\right)^n$.
Rewriting the metric in Eddington–Finkelstein coordinates, we obtain

$$ds^2 = 2\,dv\,dr - f(r)\,dv^2 + d\boldsymbol{\sigma}^2 + r^2 d\Omega_{n+1}. \tag{12}$$

We can now boost the black string along its worldvolume coordinates:

$$ds^2 = \left(\eta_{\mu\nu} + \left(\frac{r_0}{r}\right)^n \boldsymbol{u}_\mu \boldsymbol{u}_\nu\right) d\boldsymbol{\sigma}^\mu d\boldsymbol{\sigma}^\nu - 2\boldsymbol{u}_\mu d\boldsymbol{\sigma}^\mu dr + r^2 d\Omega_{n+1}, \tag{13}$$

where the worldvolume coordinates are $\sigma^\mu = (t, \sigma)$. Taking $D \to \infty$ leads to $n \to \infty$. It is then important to determine how the different quantities in the theory scale with n. The energy density of the string, $\boldsymbol{\rho} = (n+1) r_0^n$, and the pressure $\boldsymbol{P}$ are related through the following equation of state[8]

$$\boldsymbol{P} = -\frac{\boldsymbol{\rho}}{n+1}, \tag{14}$$

so the speed of sound of the perturbations is

$$c_s = \sqrt{\frac{\partial \boldsymbol{P}}{\partial \boldsymbol{\rho}}} = \sqrt{-\frac{1}{n+1}}. \tag{15}$$

This dictates the following rescaling of the coordinates and velocities:

$$\boldsymbol{\sigma}^\mu = \left(t, \frac{\sigma}{\sqrt{n}}\right), \qquad \boldsymbol{u}_\mu = \left(-1 + \mathcal{O}\left(\frac{1}{n}\right), \frac{u_\sigma}{\sqrt{n}}\right). \tag{16}$$

For convenience we define $R = r^n$ and $R_0 = r_0^n$ so that the metric becomes

$$\begin{aligned} ds^2 = &-\left(1 - \frac{R_0}{R}\right) dt^2 + 2\left(dt - \frac{1}{n} u_\sigma\, d\sigma\right) dr \\ &- 2\frac{1}{n}\frac{R_0}{R} u_\sigma\, dt\, d\sigma + \frac{1}{n}\left(1 + \frac{1}{n}\frac{R_0}{R} u_\sigma^2\right) d\sigma^2 + r^2 d\Omega_{n+1}. \end{aligned} \tag{17}$$

It is then natural to consider a metric ansatz in the form of a power-series expansion[7],

$$ds^2 = -A\,dt^2 - 2\big(u_t\,dt + u_\sigma\,d\sigma\big)dR - 2C_\sigma\,d\sigma\,dt + G_{\sigma\sigma}\,d\sigma^2. \tag{18}$$

Solving Einstein's equations order by order, it is found that the collective fields for mass density $\rho(t, \sigma)$ and momentum density $p(t, \sigma)$ must satisfy the following effective field equations

$$\partial_t \rho - \partial_\sigma^2 \rho + \partial_\sigma p = 0, \tag{19}$$

$$\partial_t p - \partial_\sigma^2 p = \partial_\sigma \left(\rho - \frac{p^2}{\rho}\right). \tag{20}$$

The detailed derivation of these equations can be found in Appendix B.

Now we will solve our effective field equations under a perturbation of the form $p(0, \sigma) = 0$, $\rho(0, \sigma) = 1 + \epsilon \cos\left(\frac{2\pi\sigma}{L}\right)$ with parameters $\varepsilon = 0.01$, $k = 0.98$, $L = \frac{2\pi}{k}$. The simulation in Fig. 1 exhibits a perturbation that

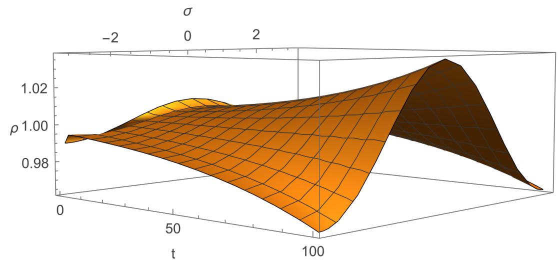


FIG. 1: Evolution of the black string mass density showing the growth of the instability.

instead of decaying grows in time, this is the defining feature of the Gregory-Laflamme (GL) instability[9].

The choice of parameters allows us to clearly observe both the exponential growth of the unstable mode and the onset of the nonlinear regime, the so-called non-uniform black string (NUBS). Finally, this instability can also be understood from Eq. (15): since $c_s^2 < 0$, the speed of sound is imaginary. This implies that perturbations do not propagate as oscillatory sound waves, but instead grow exponentially in time. Therefore, these modes are not stable propagating excitations but unstable modes, which is precisely the origin of the GL instability.

The effective theory for black strings can be easily generalized for $p \geq 1$, the collective fields $p_i(t, \sigma^a)$ and $\rho(t, \sigma^a)$ will satisfy[8]

$$\partial_t \rho - \nabla^2 \rho + \nabla_a p^a = 0, \tag{21}$$

$$\partial_t p_b - \nabla^2 p_b = \nabla_b \rho - \nabla_a \left(\frac{p_b p^a}{\rho}\right), \tag{22}$$

where the spatial brane indices $a, b = 1, \ldots p$ are raised and lowered with the flat metric δ_{ab}.

IV. BLACK HOLE SOLUTIONS AND THEIR STABILITY

In the following we specialize to $p = 2$ and use polar coordinates (r, ϕ). The explicit polar form of these equations is given in Appendix C.

Let us begin by considering configurations independent of t and ϕ. As shown explicitly in Appendix D, the equation for the radial component of the momentum can be rewritten as

$$R^{''} + \frac{R^{'}}{r} + \frac{1}{2}(R^{'})^2 + R + \frac{\Omega^2 r^2}{2} = 0, \tag{23}$$

with $R(r) = \ln \rho(r)$ and Ω being the angular velocity at the horizon. This is the master stationary equation for an axisymmetric profile. The independent term of the equation $\sim \Omega^2 r^2$ is quadratic in r, it is then natural to propose an ansatz that is quadratic in r[10]

$$R(r) = A + Br^2. \tag{24}$$

Substituting in the master equation, we find $B = \frac{-1\pm\sqrt{1-4\Omega^2}}{4}$ and $A = -4B$. Writing $\Omega = \frac{a}{1+a^2}$ and choosing the negative sign in the quadratic equation, we arrive at the following solution

$$R(r) = \frac{2}{1+a^2}\left(1 - \frac{r^2}{4}\right). \tag{25}$$

Therefore, our mass density follows a Gaussian profile

$$\rho = \exp\left[\frac{2}{1+a^2}\left(1 - \frac{r^2}{4}\right)\right] \tag{26}$$

and so do the momentum density components

$$p_r = \exp\left[\frac{2}{1+a^2}\left(1 - \frac{r^2}{4}\right)\right]\frac{-r}{1+a^2}, \tag{27}$$

$$p_\phi = \exp\left[\frac{2}{1+a^2}\left(1 - \frac{r^2}{4}\right)\right]\frac{ar^2}{1+a^2}. \tag{28}$$

We can now calculate the angular momentum per unit mass

$$\frac{J}{\rho} = \frac{\int_0^\infty drr p_\phi(r)}{\int_0^\infty drr\rho(r)} = 2a, \tag{29}$$

that reproduces correctly the value at large D for the Myers-Perry black hole and allows us to identify the axisymmetric solution we have found with the Myers-Perry black hole solution[10]. The limit $a \to 0$ is the so-called static limit, whereas the limit $a \to \infty$ for which $\frac{J}{\rho} \to \infty$ is the ultra-spinning limit. Had we chosen the positive solution when solving the quadratic equation we would have obtained $\frac{J}{\rho} = \frac{2}{a}$ and the two limits would be inverted.

Lastly let us boost our Myers-Perry solution $\sigma_a \to \sigma_a - u_a t$, in Cartesian coordinates (x^1, x^2) we obtain

$$\rho(t, x) = \exp\left[\frac{2}{1+a^2}\left(1 - \frac{(x_i - u_i t)(x^i - u^i t)}{4}\right)\right]. \tag{30}$$

Now let us consider a stationary configuration that is not axisymmetric nor time-independent, and taking a purely rotational velocity field $v^\phi = \Omega$ the equation for $R = R(r, \phi - \Omega t)$ becomes

$$\begin{aligned}0 = \partial_r^2 R + \frac{\partial_r R}{r} + \frac{\partial_\phi^2 R}{r^2} + \frac{1}{2}\left((\partial_r R)^2 + \frac{(\partial_\phi R)^2}{r^2}\right) \\ + R + \frac{\Omega^2 r^2}{2},\end{aligned} \tag{31}$$

which reduces to Eq. (23) for axisymmetric time-independent profiles. The mass density is

$$\rho = e^{R(r,\phi-\Omega t)} \tag{32}$$

and the momenta are

$$p_r = \partial_r \rho, \qquad p_\phi = \partial_\phi \rho + \Omega r^2 \rho. \tag{33}$$

After a long computation, which is done in Appendix E, we can solve this equation to obtain

$$R(t, r, \phi) = 1 - \frac{r^2}{4}\left(1 + \sqrt{1 - 4\Omega^2}\cos(2(\phi - \Omega t))\right) \tag{34}$$

and the corresponding mass density

$$\rho(t, r, \phi) = \exp\left[1 - \frac{r^2}{4}(1 + \sqrt{1 - 4\Omega^2}\cos(2(\phi - \Omega t)))\right] \tag{35}$$

that describes a rotating bar. It is then useful to rewrite our solution in rotating Cartesian coordinates.

$$x = x^1 \cos\Omega t + x^2 \sin\Omega t, \qquad y = x^2 \cos\Omega t - x^1 \sin\Omega t. \tag{36}$$

Defining the length $l_\parallel$ and width $l_\perp$ of the bar as follows[10]

$$l_\parallel^2 = \frac{2}{1 - \sqrt{1 - 4\Omega^2}}, \qquad l_\perp^2 = \frac{2}{1 + \sqrt{1 - 4\Omega^2}}. \tag{37}$$

The solution then reads

$$R(x, y) = 1 - \frac{x^2}{2l_\perp^2} - \frac{y^2}{2l_\parallel^2}. \tag{38}$$

The angular momentum per unit mass for this solution is

$$\frac{J}{\rho} = \frac{1}{\Omega}. \tag{39}$$

Now we want to study small linearized perturbations of the black hole solutions we have found. We start by adding a perturbation to the Myers-Perry solution[10]

$$R(r) = R_0(r) + \varepsilon \delta R(r) e^{i\rho_\phi(\phi - \Omega t)}, \tag{40}$$

where ε is an infinitesimal parameter. When $\rho_\phi = 0$ the perturbations are time-independent and axisymmetric and therefore they are zero modes.

Zero modes are particularly interesting for they lead to new branches of stationary solutions.
When $\rho_\phi \neq 0$ the frequency of the perturbation is $\omega = \rho_\phi \Omega$, in this case the perturbation corotates with the black hole and therefore we also consider them zero modes. The explicit derivation of the zero-mode conditions used below is given in Appendix F. We will have the corotating zero modes for

$$a_c^2 = |\rho_\phi| + 2k - 1 \quad , k = 0, 1, 2, ... \tag{41}$$

with $k = \frac{a^2+1-|\rho_\phi|}{2}$. Considering $|\rho_\phi| > 0$ we have zero modes for

$$a = \sqrt{1}, \sqrt{2}, \sqrt{3} \to \Omega = \frac{1}{2}, \frac{\sqrt{2}}{3}, \frac{\sqrt{3}}{4}. \tag{42}$$

In the axisymmetric time-independent case we will have zero modes for

$$a_c^2 = 2k - 1 \quad , k = 2, 3. \tag{43}$$

In this case we have

$$a = \sqrt{3}, \sqrt{5} \to \Omega = \frac{\sqrt{3}}{4}, \frac{\sqrt{5}}{6}. \tag{44}$$

We can thus see that the first zero mode from which a new branch of solutions appears is obtained for $a = 1 \to \Omega = \frac{1}{2} \to \frac{J}{\rho} = 2$. From this point a branch of solutions corresponding to black bars appears and the Myers-Perry branch becomes metastable.
Now let us consider corotating perturbations for black bar solutions. Starting from the black bar solution in Cartesian coordinates and adding a perturbation[10]

$$R(x, y) = R_0(x, y) + \varepsilon \delta R e^{i\rho_\phi(\phi - \Omega t)}, \tag{45}$$

we find that the zero modes occur for

$$\Omega = \frac{\sqrt{n_y - 1}}{n_y}. \tag{46}$$

For $n_y = 2$ we recover a bar mode of the Myers-Perry black hole with $\Omega = \frac{1}{2}$, thus the new zero modes are

$$n_y = 3, 4, 5 \to \Omega = \frac{\sqrt{2}}{3}, \frac{\sqrt{3}}{4}, \frac{\sqrt{4}}{5}. \tag{47}$$

The modes we are interested in are the even modes $n_y = 4, 6, 8$ which lead to symmetrical deformations around the center of the bar. From $n_y = 4 \to \Omega = \frac{\sqrt{3}}{4} \to \frac{J}{\rho} = \frac{4}{\sqrt{3}}$ we obtain a new branch of solutions resembling dumbbells, and the black bar branch becomes metastable.
We now study the stability of the dumbbell branch. Since no analytic expression is known for this branch, we reconstruct it numerically using a shooting method[11], the numerical construction is detailed in Appendix G:1. We reduce the stationary problem to an ordinary differential equation for the deformation $u(y)$ of the black bar. The initial value $u_0 = u(0)$ is used as the shooting parameter i.e, for each value of Ω we choose u_0 so that the solution remains regular. The resulting system is integrated with an adaptive Runge-Kutta, the solution obtained for one value of Ω is then used as a predictor in order to continue the branch. The resulting branch reaches a turning point where J/ρ starts to decrease, as shown in Fig. 2. Beyond this point, the numerical solution eventually returns to the black bar branch. The point where J/ρ starts to decrease is a Poincaré turning point of the dumbbell branch, signaling the appearance of an additional negative mode and hence dynamical instability[12]. Numerically, this occurs at $\left(\frac{J}{\rho}\right)_c = 2.66202712$. Above this threshold, no stable black hole solution of the effective equations exists, suggesting that perturbations grow exponentially and may lead to a violation of cosmic censorship.

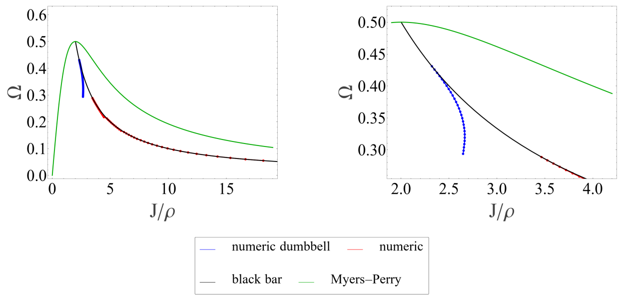
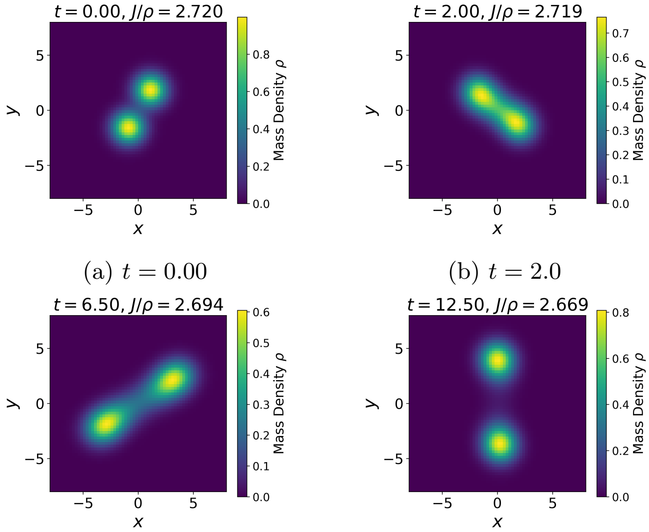


FIG. 2: Phase diagram of the Myers–Perry, black bar, and dumbbell branches, along with a zoom-in of the dumbbell branch.

V. BLACK HOLE COLLISIONS AND COSMIC CENSORSHIP

The violation of WCC, through the formation of a naked singularity, signals the breakdown of classical General Relativity and the need for quantum gravity. Such processes typically drive low-energy configurations into regimes where quantum effects become relevant, often requiring fine-tuned initial conditions[3]. The previous sections have shown two important ingredients for this mechanism. First, black strings are unstable under long-wavelength perturbations. Second, in the large-D effective theory there exists a critical value of the angular momentum above which no stable stationary black hole solution is available. A natural setting for such large angular momentum configurations is the collision of two black holes, where the large-D limit provides a controlled framework to study whether the post-merger state develops a neck-thinning instability[3].
Solving numerically Eqs. (21) and (22) for a system of two boosted black holes with initial angular momentum $\frac{J}{\rho} = 2.72$ allows us to simulate the collision, merger, and breakup leading to the WCC violation. The time evolution is performed on a Cartesian grid. The spatial deriva-

tives are computed using fast Fourier transforms(FFTs) and the resulting system of ordinary differential equations is solved for its time evolution with a fourth-order Runge-Kutta method, the numerical construction is detailed in Appendix G:2. The results shown in Fig. 3 provide evidence that, at very large D, black hole collisions with $J/\rho > 2.66$ form a bar-like horizon whose central neck progressively shrinks. In the strict $D \to \infty$ limit, the Gaussian profiles prevent the neck from reaching exactly zero size, so the naked singularity is not directly observed in the simulation[4]. However, by analogy with the Gregory–Laflamme instability and its finite-D nonlinear evolution, this neck-thinning process is expected to continue at finite D, eventually leading to horizon pinch-off and the formation of a naked singularity[5].



(a) $t = 0.00$ (b) $t = 2.0$ (c) $t = 6.50$ (d) $t = 12.50$

FIG. 3: Evolution of a high-angular-momentum black hole collision.

VI. CONCLUSIONS

In this work the large-D limit of General Relativity has been used as an effective framework to explore instances of WCC violation in higher-dimensional black hole dynamics. Firstly we performed a dimensional reduction in order to work properly with spacetimes of arbitrary number of dimensions. We then applied this formalism to black strings to obtain the effective equations and by perturbing them we recovered the GL instability.
Then we studied the effective theory for 2-branes and found two types of stationary rotating solutions: the axisymmetric Myers-Perry black holes and the non-axisymmetric black bars. We analyzed their perturbations and identified the zero modes from which new branches of stationary solutions appeared.
We then studied the dumbbell branch through numerical methods and found that the branch becomes unstable when the angular momentum per unit mass reaches the critical value $\frac{J}{\rho} \approx 2.662$. Finally, the collision of two boosted black holes with total angular momentum above the critical value forms an elongated bar-like state whose central neck progressively shrinks. Although in the strict $D \to \infty$ limit the Gaussian profiles prevent complete pinch-off, by analogy with the Gregory–Laflamme instability this process is expected to continue at finite but large D, leading to a naked singularity and therefore to a violation of WCC.

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Violació de Censura Còsmica a gran D

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Resum: En aquest treball, el límit de gran D de la Relativitat General s'utilitza com a marc per investigar possibles violacions de la censura còsmica feble i en darrer terme reproduïr els resultats de [3, 4]. En aquest límit, les equacions d'Einstein se simplifiquen i permeten descriure la dinàmica gravitatòria en termes de camps col·lectius. Es duu a terme una reducció dimensional per tal de treballar amb espaitemps d'un nombre arbitrari de dimensions. Estudiem l'estabilitat de diferents objectes en aquest límit, com ara les cordes negres i les barres negres. Les simulacions numèriques mostren que, per a un moment angular per unitat de massa prou elevat, no hi ha solucions estacionàries estables de forats negres en la teoria. Finalment, proporcionem evidències d'una violació de la censura còsmica feble mitjançant la simulació de la col·lisió de dos forats negres sota un boost.
Paraules clau: Gravitació, forats negres, hipòtesi de censura còsmica
ODSs: Educació de qualitat

Objectius de Desenvolupament Sostenible (ODSs o SDGs)

1. Fi de les desigualtats		10. Reducció de les desigualtats	
2. Fam zero		11. Ciutats i comunitats sostenibles	
3. Salut i benestar		12. Consum i producció responsables	
4. Educació de qualitat	X	13. Acció climàtica	
5. Igualtat de gènere		14. Vida submarina	
6. Aigua neta i sanejament		15. Vida terrestre	
7. Energia neta i sostenible		16. Pau, justícia i institucions sòlides	
8. Treball digne i creixement econòmic		17. Aliança pels objectius	
9. Indústria, innovació, infraestructures			

Appendix A: Dimensional Reduction Computations

We will start by computing the Christoffel symbols, in general

$$\Gamma^k_{ij} = \frac{1}{2} g^{kr}(\partial_i g_{jr} + \partial_j g_{ir} - \partial_r g_{ij}) \tag{A1}$$

The Christoffel symbols for the brane

$$\bar{\Gamma}^\rho_{\mu\nu} = \frac{1}{2} g^{\rho\sigma}(\partial_\mu g_{\nu\sigma} + \partial_\nu g_{\sigma\mu} - \partial_\sigma g_{\mu\nu}) = \Gamma^\rho_{\mu\nu}(x) \tag{A2}$$

and the purely spherical terms

$$\bar{\Gamma}^a_{bc} = \frac{1}{2} g^{ad}(\partial_b g_{cd} + \partial_c g_{bd} - \partial_d g_{bc}) = \Gamma^a_{bc}(\gamma) \tag{A3}$$

We now focus on the mixed Christoffel symbols, those with Latin and Greek indices.

$$\bar{\Gamma}^a_{\mu\nu} = \frac{1}{2} g^{ab}(\partial_\mu g_{\nu b} + \partial_\nu g_{\mu b} - \partial_b g_{\mu\nu}) = 0 \tag{A4}$$

where the first two terms are equal to 0 because the metric contains no mixed terms, and the third term is equal to 0 because $g_{\mu\nu} = g_{\mu\nu}(x)$. By analogous arguments we can also compute

$$\bar{\Gamma}^\nu_{\mu a} = \frac{1}{2} g^{\nu\rho}(\partial_\mu g_{\rho a} + \partial_a g_{\mu\rho} - \partial_\rho g_{\mu a}) = 0 \tag{A5}$$

This follows from the block-diagonal structure of the metric, which implies that there are no mixed components between the brane and the spherical directions. We now compute the nonzero mixed Christoffel symbols

$$\begin{aligned}\bar{\Gamma}^\mu_{ab} &= \frac{1}{2} g^{\mu\rho}(\partial_a g_{\rho b} + \partial_b g_{a\rho} - \partial_\rho g_{ab}) \\ &= -\frac{1}{2} g^{\mu\rho}\partial_\rho(e^{2\phi}\gamma_{ab}) = -g^{\mu\rho}\gamma_{ab}e^{2\phi}\partial_\rho\phi,\end{aligned} \tag{A6}$$

$$\begin{aligned}\bar{\Gamma}^b_{a\mu} &= \frac{1}{2} g^{bc}(\partial_a g_{c\mu} + \partial_\mu g_{ac} - \partial_c g_{a\mu}) \\ &= \frac{1}{2} g^{bc}\partial_\mu g_{ac} = \frac{1}{2}\gamma^{bc}e^{-2\phi}\partial_\mu(e^{2\phi}\gamma_{ac}) \\ &= \gamma^{bc}\gamma_{ac}\,\partial_\mu\phi = \delta^b_a\,\partial_\mu\phi.\end{aligned} \tag{A7}$$

We can now compute the Riemann tensor, in general

$$R_{ijk}{}^l = \partial_j\Gamma^l_{ik} - \partial_i\Gamma^l_{jk} + \Gamma^r_{ik}\Gamma^l_{jr} - \Gamma^r_{jk}\Gamma^l_{ir} \tag{A8}$$

For the components of the brane

$$\begin{aligned}\bar{R}_{\mu\nu\rho}{}^\sigma &= \partial_\nu\bar{\Gamma}^\sigma_{\mu\rho} - \partial_\mu\bar{\Gamma}^\sigma_{\nu\rho} + \bar{\Gamma}^\epsilon_{\mu\rho}\bar{\Gamma}^\sigma_{\nu\epsilon} - \bar{\Gamma}^\epsilon_{\nu\rho}\bar{\Gamma}^\sigma_{\mu\epsilon} \\ &\quad + \bar{\Gamma}^a_{\mu\rho}\bar{\Gamma}^\sigma_{\nu a} - \bar{\Gamma}^a_{\nu\rho}\bar{\Gamma}^\sigma_{\mu a} = R_{\mu\nu\rho}{}^\sigma.\end{aligned} \tag{A9}$$

For the spherical components

$$\begin{aligned}\bar{R}_{abc}{}^d &= \partial_b\bar{\Gamma}^d_{ac} - \partial_a\bar{\Gamma}^d_{bc} + \bar{\Gamma}^e_{ac}\bar{\Gamma}^d_{be} - \bar{\Gamma}^e_{bc}\bar{\Gamma}^d_{ae} \\ &\quad + \bar{\Gamma}^\mu_{ac}\bar{\Gamma}^d_{b\mu} - \bar{\Gamma}^\mu_{bc}\bar{\Gamma}^d_{a\mu} \\ &= R_{abc}{}^d + \bar{\Gamma}^\mu_{ac}\bar{\Gamma}^d_{b\mu} - \bar{\Gamma}^\mu_{bc}\bar{\Gamma}^d_{a\mu} \\ &= R_{abc}{}^d + \big(-g^{\mu\rho}\gamma_{ac}e^{2\phi}\partial_\rho\phi\big)\big(\delta^d_b\partial_\mu\phi\big) \\ &\quad - \big(-g^{\mu\rho}\gamma_{bc}e^{2\phi}\partial_\rho\phi\big)\big(\delta^d_a\partial_\mu\phi\big) \\ &= R_{abc}{}^d + g^{\mu\rho}e^{2\phi}(\nabla_\rho\phi)(\nabla_\mu\phi)\big[\gamma_{bc}\delta^d_a - \gamma_{ac}\delta^d_b\big].\end{aligned} \tag{A10}$$

where in the last step we have used that ϕ is a scalar and therefore $\partial_\mu\phi = \nabla_\mu\phi$. Now let us compute the mixed terms

$$\begin{aligned}\bar{R}^a_{\mu\nu\rho} &= \partial_\nu\bar{\Gamma}^a_{\mu\rho} - \partial_\mu\bar{\Gamma}^a_{\nu\rho} + \bar{\Gamma}^\varepsilon_{\mu\rho}\bar{\Gamma}^a_{\nu\varepsilon} - \bar{\Gamma}^r_{\nu\rho}\bar{\Gamma}^a_{\mu r} \\ &\quad + \bar{\Gamma}^b_{\mu\rho}\bar{\Gamma}^a_{\nu b} - \bar{\Gamma}^b_{\nu\rho}\bar{\Gamma}^a_{\mu b} = 0,\end{aligned} \tag{A11}$$

$$\begin{aligned}\bar{R}^\rho_{\mu\nu a} &= \partial_\nu\bar{\Gamma}^\rho_{\mu a} - \partial_\mu\bar{\Gamma}^\rho_{\nu a} + \bar{\Gamma}^\varepsilon_{\mu a}\bar{\Gamma}^\rho_{\nu\varepsilon} - \bar{\Gamma}^\varepsilon_{\nu a}\bar{\Gamma}^\rho_{\mu\varepsilon} \\ &\quad + \bar{\Gamma}^b_{\mu a}\bar{\Gamma}^\rho_{\nu b} - \bar{\Gamma}^b_{\nu a}\bar{\Gamma}^\rho_{\mu b} = 0,\end{aligned} \tag{A12}$$

$$\begin{aligned}\bar{R}_{ab\rho}{}^c &= \partial_b\bar{\Gamma}^c_{\rho a} - \partial_a\bar{\Gamma}^c_{b\rho} + \bar{\Gamma}^\delta_{a\rho}\bar{\Gamma}^c_{b\delta} - \bar{\Gamma}^\delta_{b\rho}\bar{\Gamma}^c_{\delta a} \\ &\quad + \bar{\Gamma}^d_{a\rho}\bar{\Gamma}^c_{bd} - \bar{\Gamma}^d_{b\rho}\bar{\Gamma}^c_{da} \\ &= \partial_b\big(\delta^c_a\partial_\rho\phi\big) - \partial_a\big(\delta^c_b\partial_\rho\phi\big) \\ &\quad + \Gamma^c_{db}\big(\delta^d_a\partial_\rho\phi\big) - \Gamma^c_{da}\big(\delta^d_b\partial_\rho\phi\big) \\ &= \delta^c_a\partial_b\partial_\rho\phi - \delta^c_b\partial_a\partial_\rho\phi + \Gamma^c_{ba}\partial_\rho\phi - \Gamma^c_{ba}\partial_\rho\phi \\ &= \delta^c_a\partial_b\nabla_\rho\phi - \delta^c_b\partial_a\nabla_\rho\phi = 0,\end{aligned} \tag{A13}$$

where on the last line we have used that $\phi = \phi(x)$

$$\begin{aligned}R_{abc}{}^\rho &= \partial_b\bar{\Gamma}^\rho_{ac} - \partial_a\bar{\Gamma}^\rho_{bc} + \bar{\Gamma}^\delta_{ac}\bar{\Gamma}^\rho_{b\delta} - \bar{\Gamma}^d_{bc}\bar{\Gamma}^\rho_{ad} \\ &\quad + \bar{\Gamma}^d_{ac}\bar{\Gamma}^\rho_{bd} - \bar{\Gamma}^d_{bc}\bar{\Gamma}^\rho_{ad} \\ &= \partial_b\big(-g^{\rho\delta}\gamma_{ac}e^{2\phi}\partial_\delta\phi\big) - \partial_a\big(-g^{\rho\delta}\gamma_{bc}e^{2\phi}\partial_\delta\phi\big) \\ &\quad + \Gamma^d_{ac}\big(-g^{\rho\delta}\gamma_{bd}e^{2\phi}\partial_\delta\phi\big) - \Gamma^d_{bc}\big(-g^{\rho\delta}\gamma_{ad}e^{2\phi}\partial_\delta\phi\big) \\ &= -g^{\rho\delta}e^{2\phi}\partial_\delta\phi\big[\partial_b\gamma_{ac} + \Gamma^d_{ac}\gamma_{bd} - \partial_a\gamma_{bc} + \Gamma^d_{bc}\gamma_{ad}\big] \\ &= -g^{\rho\delta}e^{2\phi}\partial_\delta\phi\big[\nabla_b\gamma_{ac} - \nabla_a\gamma_{bc}\big] = 0,\end{aligned} \tag{A14}$$

where we have used that the covariant derivative of a rank-2 covariant tensor is $\nabla_a T_{bc} = \partial_a T_{bc} - \Gamma^d_{ab}T_{dc} - \Gamma^d_{ac}T_{bd}$ and the fact that the metric is covariantly constant. We now compute the nonzero mixed components

$$\begin{aligned}\bar{R}_{\mu a\nu}{}^b &= \partial_a\bar{\Gamma}^b_{\mu\nu} - \partial_\mu\bar{\Gamma}^b_{a\nu} + \bar{\Gamma}^\varepsilon_{\mu\nu}\bar{\Gamma}^b_{a\varepsilon} - \bar{\Gamma}^\varepsilon_{a\nu}\bar{\Gamma}^b_{\mu\varepsilon} \\ &\quad + \bar{\Gamma}^c_{\mu\nu}\bar{\Gamma}^b_{ac} - \bar{\Gamma}^c_{a\nu}\bar{\Gamma}^b_{\mu c} \\ &= -\delta^b_a\partial_\mu\partial_\nu\phi + \Gamma^\varepsilon_{\mu\nu}\delta^b_a\partial_\varepsilon\phi - \delta^b_a\partial_\nu\phi\partial_\mu\phi \\ &= -\delta^b_a\big(\partial_\mu\partial_\nu\phi - \Gamma^\varepsilon_{\mu\nu}\partial_\varepsilon\phi + \partial_\nu\phi\partial_\mu\phi\big) \\ &= -\delta^b_a\big(\nabla_\mu\nabla_\nu\phi + \nabla_\nu\phi\nabla_\mu\phi\big),\end{aligned} \tag{A15}$$

where on the last step we have used, that for a scalar $\nabla_\mu\nabla_\nu\phi = \partial_\mu\partial_\nu\phi - \Gamma^\varepsilon_{\mu\nu}\partial_\varepsilon\phi$.

$$\begin{aligned}\bar{R}_{a\mu b}{}^\nu &= \partial_\mu\bar{\Gamma}^\nu_{ab} - \partial_a\bar{\Gamma}^\nu_{\mu b} + \bar{\Gamma}^\varepsilon_{ab}\bar{\Gamma}^\nu_{\mu\varepsilon} - \bar{\Gamma}^\varepsilon_{\mu b}\bar{\Gamma}^\nu_{a\varepsilon} \\ &\quad + \bar{\Gamma}^c_{ab}\bar{\Gamma}^\nu_{c\mu} - \bar{\Gamma}^c_{\mu b}\bar{\Gamma}^\nu_{ac} \\ &= -\partial_\mu(g^{\nu\rho}\gamma_{ab}e^{2\phi}\partial_\rho\phi) - \Gamma^\nu_{\varepsilon\mu}(g^{\varepsilon\rho}\gamma_{ab}e^{2\phi}\partial_\rho\phi) \\ &\quad + \delta^c_b\gamma_{ac}g^{\nu\rho}e^{2\phi}\partial_\mu\phi\partial_\rho\phi \\ &= -\nabla_\mu\big(g^{\nu\rho}\gamma_{ab}e^{2\phi}\nabla_\rho\phi\big) + g^{\nu\rho}e^{2\phi}\gamma_{ab}\nabla_\mu\phi\nabla_\rho\phi \\ &= g^{\nu\rho}\gamma_{ab}e^{2\phi}\big(\nabla_\mu\phi\nabla_\rho\phi + \nabla_\mu\nabla_\rho\phi\big).\end{aligned} \tag{A16}$$

Finally, we compute the Ricci tensor. It is immediate to see that the mixed components vanish. For the spherical components we obtain

$$\begin{aligned}\bar{R}_{ab} &= \bar{R}^{c}{}_{abc} + \bar{R}^{\nu}{}_{ab\nu} \\ &= R_{ab} + \big[\gamma_{bc}\delta^c_a - \gamma_{ab}\delta^c_c\big] g^{\mu\rho} e^{2\phi}\nabla_\rho\phi\nabla_\mu\phi \\ &\quad + g^{\mu\rho}\gamma_{ab}e^{2\phi}\big[\nabla_\mu\phi\nabla_\rho\phi + \nabla_\mu\nabla_\rho\phi\big] \\ &= R_{ab} - g^{\mu\rho}\gamma_{ab}e^{2\phi}\big[\nabla_\mu\nabla_\rho\phi + (n+1)\nabla_\mu\phi\nabla_\rho\phi\big].\end{aligned} \tag{A17}$$

For the brane components we find

$$\begin{aligned}\bar{R}_{\mu\nu} &= \bar{R}^{\rho}{}_{\mu\nu\rho} + \bar{R}^{a}_{\mu\nu a} \\ &= R_{\mu\nu} - (n+1)\big(\nabla_\mu\nabla_\nu\phi + \nabla_\nu\phi\nabla_\mu\phi\big).\end{aligned} \tag{A18}$$

Appendix B: Explicit Derivation of the effective equations

Starting from the metric ansatz

$$ds^2 = -A\,dt^2 - 2\big(u_t\,dt + u_\sigma\,d\sigma\big)dR - 2C_\sigma\,d\sigma\,dt + G_{\sigma\sigma}\,d\sigma^2, \tag{B1}$$

with

$$A(t,\sigma,R) = \sum_{k=0}^{k_{\max}} \frac{1}{n^k} A_k(t,\sigma,R) + O\left(\frac{1}{n^{k_{\max}+1}}\right), \tag{B2}$$

$$u_t(t,\sigma,R) = \sum_{k=0}^{k_{\max}} \frac{1}{n^k} u_{t,k}(t,\sigma,R) + O\left(\frac{1}{n^{k_{\max}+1}}\right), \tag{B3}$$

$$u_\sigma(t,\sigma,R) = \sum_{k=0}^{k_{\max}} \frac{1}{n^{k+1}} u_{\sigma,k}(t,\sigma,R) + O\left(\frac{1}{n^{k_{\max}+2}}\right), \tag{B4}$$

$$C_\sigma(t,\sigma,R) = \sum_{k=0}^{k_{\max}} \frac{1}{n^{k+1}} C_{\sigma,k}(t,\sigma,R) + O\left(\frac{1}{n^{k_{\max}+2}}\right), \tag{B5}$$

$$G_{\sigma\sigma}(t,\sigma,R) = \frac{1}{n} + \sum_{k=0}^{k_{\max}} \frac{1}{n^{k+1}} G_{\sigma\sigma,k}(t,\sigma,R) + O\left(\frac{1}{n^{k_{\max}+3}}\right). \tag{B6}$$

We fix the gauge by taking $u_{t,0} = -1$, $u_{\sigma,0} = u_\sigma$ At leading order, the Einstein equations arising from the black string metric ansatz together with the warp factor reduce to

$$R_{tR} = 0 \implies \partial_R A_0 + \frac{R}{2}\,\partial_R^2 A_0 = 0, \tag{B7}$$

$$R_{R\sigma} = 0 \implies \partial_R C_{\sigma,0} + \frac{R}{2}\,\partial_R^2 C_{\sigma,0} = 0, \tag{B8}$$

$$\begin{aligned}R_{\sigma\sigma} = 0 \implies \partial_\sigma C_{\sigma,0} &+ \frac{R}{2}\Big[R\big(\partial_R C_{\sigma,0}\big)^2 \\ &+ \big(2A_0 + R\,\partial_R A_0\big)\partial_R G_{\sigma\sigma,0} + R\,A_0\,\partial_R^2 G_{\sigma\sigma,0} \\ &+ 2\,\partial_\sigma\partial_R C_{\sigma,0}\Big] = 0.\end{aligned} \tag{B9}$$

Solving the differential equations for A_0 and $C_{\sigma,0}$, we find

$$A_0(t,\sigma,R) = -\frac{C_1(t,\sigma)}{R} + C_2(t,\sigma), \tag{B10}$$

$$C_{\sigma,0}(t,\sigma,R) = -\frac{C_3(t,\sigma)}{R} + C_4(t,\sigma). \tag{B11}$$

Imposing regularity at the horizon and asymptotic flatness, we obtain

$$C_2(t,\sigma) = 1, \qquad C_1(t,\sigma) = \rho(t,\sigma), \tag{B12}$$

$$C_4(t,\sigma) = 0, \qquad C_3(t,\sigma) = -p(t,\sigma), \tag{B13}$$

so that

$$A_0(t,\sigma,R) = 1 - \frac{\rho(t,\sigma)}{R}, \tag{B14}$$

$$C_{\sigma,0}(t,\sigma,R) = \frac{p(t,\sigma)}{R}. \tag{B15}$$

Substituting these expressions into the equation for $R_{\sigma\sigma} = 0$, we obtain

$$\begin{aligned}G_{\sigma\sigma,0}(t,\sigma,R) = C_5(t,\sigma) &+ \frac{p(t,\sigma)^2}{\rho(t,\sigma)^2}\log\left(1 - \frac{\rho(t,\sigma)}{R}\right) \\ &+ \frac{R\,C_6(t,\sigma)\,\log\left(1 - \frac{\rho(t,\sigma)}{R}\right) + p(t,\sigma)^2}{R\,\rho(t,\sigma)}.\end{aligned} \tag{B16}$$

Imposing regularity at the horizon once again, the solution is

$$G_{\sigma\sigma,0}(t,\sigma,R) = \frac{p(t,\sigma)^2}{R\,\rho(t,\sigma)}. \tag{B17}$$

At next-to-leading order we keep $k_{\max} = 1$. The metric functions are expanded as

$$A = A_0 + \frac{1}{n}A_1 + \mathcal{O}(n^{-2}), \tag{B18}$$

$$u_t = u_{t,0} + \frac{1}{n}u_{t,1} + \mathcal{O}(n^{-2}), \tag{B19}$$

$$u_\sigma = \frac{1}{n}u_{\sigma,0} + \frac{1}{n^2}u_{\sigma,1} + \mathcal{O}(n^{-3}), \tag{B20}$$

$$C_\sigma = \frac{1}{n}C_{\sigma,0} + \frac{1}{n^2}C_{\sigma,1} + \mathcal{O}(n^{-3}), \tag{B21}$$

$$G_{\sigma\sigma} = \frac{1}{n}\left[1 + \frac{1}{n}G_{\sigma\sigma,0} + \frac{1}{n^2}G_{\sigma\sigma,1} + \mathcal{O}(n^{-3})\right]. \tag{B22}$$

where all functions depend on (t,σ,R). We keep the gauge choice

$$u_{t,0} = -1, \qquad u_{\sigma,0} = u_\sigma. \tag{B23}$$

Using Eqs. (B14), (B15) and (B17), the tt component of the Ricci tensor at the next leading order gives

$$R^{(1)}_{tt} = \frac{1}{2R}\left(\partial_\sigma p - \partial_\sigma^2\rho + \partial_t\rho\right). \tag{B24}$$

The vacuum equation $R_{tt}^{(1)} = 0$ therefore gives

$$\partial_t \rho - \partial_\sigma^2 \rho + \partial_\sigma p = 0. \tag{B25}$$

The second effective equation follows from the mixed component $R_{t\sigma}$. At the corresponding order one obtains

$$R_{t\sigma}^{(1)} = \frac{\mathcal{E}_p}{2R\rho^2}, \tag{B26}$$

with

$$\begin{aligned}\mathcal{E}_p = p^2 \partial_\sigma \rho - 2\rho p\, \partial_\sigma p \\ + \rho^2 \left(\partial_\sigma \rho + \partial_\sigma^2 p - \partial_t p\right).\end{aligned} \tag{B27}$$

The vacuum equation $R_{t\sigma}^{(1)} = 0$ implies

$$\mathcal{E}_p = 0. \tag{B28}$$

Dividing Eq. (B28) by ρ^2, we find

$$\begin{aligned}0 = \partial_\sigma \rho + \partial_\sigma^2 p - \partial_t p \\ + \frac{p^2}{\rho^2}\partial_\sigma \rho - 2\frac{p}{\rho}\partial_\sigma p.\end{aligned} \tag{B29}$$

Using the identity

$$\partial_\sigma \left(\frac{p^2}{\rho}\right) = 2\frac{p}{\rho}\partial_\sigma p - \frac{p^2}{\rho^2}\partial_\sigma \rho, \tag{B30}$$

Eq. (B29) becomes

$$\partial_t p - \partial_\sigma^2 p = \partial_\sigma \rho - \partial_\sigma \left(\frac{p^2}{\rho}\right). \tag{B31}$$

Equivalently,

$$\partial_t p - \partial_\sigma^2 p = \partial_\sigma \left(\rho - \frac{p^2}{\rho}\right). \tag{B32}$$

Thus, the next-to-leading order Einstein equations impose the effective large-D equations

$$\partial_t \rho - \partial_\sigma^2 \rho + \partial_\sigma p = 0 \tag{B33}$$

and

$$\partial_t p - \partial_\sigma^2 p = \partial_\sigma \left(\rho - \frac{p^2}{\rho}\right) \tag{B34}$$

Appendix C: Derivation of the effective equations in polar coordinates

Let us see how can we write the effective equations in polar coordinates.

$$ds^2 = dr^2 + r^2 d\phi^2, \tag{C1}$$

$$p = p_r dr + p_\phi d\phi, \tag{C2}$$

The laplacian of ρ

$$\nabla^2 \rho = \partial_r^2 \rho + \frac{1}{r}\partial_r \rho + \frac{1}{r^2}\partial_\phi^2 \rho \tag{C3}$$

and the divergence of p

$$\nabla_a p^a = \partial_r p^r + \frac{p^r}{r} + \partial_\phi p^\phi \tag{C4}$$

substituting into the effective equation for the density

$$\partial_t \rho = \partial_r^2 \rho + \frac{1}{r}\partial_r \rho + \frac{1}{r^2}\partial_\phi^2 \rho - \partial_r p^r - \frac{p^r}{r} - \partial_\phi p^\phi \tag{C5}$$

considering the metric in polar coordinates $g_{ij} = diag(1, r^2)$ then $p^r = p_r$ and $p^\phi = \frac{p_\phi}{r^2}$ so we end up finding

$$\partial_t \rho = \partial_r^2 \rho + \frac{1}{r}\partial_r \rho + \frac{1}{r^2}\partial_\phi^2 \rho - \partial_r p_r - \frac{p_r}{r} - \partial_\phi \frac{p_\phi}{r^2} \tag{C6}$$

let us now focus on the equation for the momenta. For the radial component, we write $b = r$ then

$$\nabla^2 p_r = \partial_r^2 p_r + \frac{1}{r}\partial_r p_r - \frac{p_r}{r^2} + \frac{1}{r^2}\partial_\phi^2 p_r - \frac{2}{r^3}\partial_\phi p_\phi \tag{C7}$$

and

$$\begin{aligned}\nabla_a \left(\frac{p_r p^a}{\rho}\right) = \partial_r \left(\frac{p_r^2}{\rho}\right) + \frac{1}{r^2}\partial_\phi \left(\frac{p_r p_\phi}{\rho}\right) \\ + \frac{p_r^2}{r\rho} - \frac{p_\phi^2}{r^3\rho}.\end{aligned} \tag{C8}$$

substituting in the equation for the momenta

$$\begin{aligned}\partial_t p_r = \partial_r \rho + \partial_r^2 p_r + \frac{1}{r}\partial_r p_r - \frac{p_r}{r^2} + \frac{1}{r^2}\partial_\phi^2 p_r - \frac{2}{r^3}\partial_\phi p_\phi \\ - \partial_r \left(\frac{p_r^2}{\rho}\right) - \frac{1}{r^2}\partial_\phi \left(\frac{p_r p_\phi}{\rho}\right) - \frac{p_r^2}{r\rho} + \frac{p_\phi^2}{r^3\rho}.\end{aligned} \tag{C9}$$

now, for the angular component, putting $b = \phi$ and proceeding in an analogous fashion

$$\begin{aligned}\partial_t p_\phi = \partial_\phi \rho + \frac{\partial_\phi^2 p_\phi}{r^2} - \frac{\partial_r p_\phi}{r} + \partial_r^2 p_\phi + \frac{2}{r}\partial_\phi p_r \\ - \frac{1}{r^2}\partial_\phi^2 \left(\frac{p_\phi^2}{\rho}\right) - \partial_r \left(\frac{p_r p_\phi}{\rho}\right) - \frac{p_r p_\phi}{r\rho}.\end{aligned} \tag{C10}$$

Appendix D: Explicit Derivation of the Myers-Perry black hole solution

The density equation for a configuration independent of t and ϕ is

$$\partial_r^2 \rho + \frac{1}{r}\partial_r \rho - \partial_r p_r - \frac{p_r}{r} = 0 \tag{D1}$$

which can be rewritten as

$$\frac{1}{r}\partial_r (r\partial_r \rho) = \frac{1}{r}\partial_r (r p_r) \tag{D2}$$

which can be easily solved by integrating once and imposing regularity at the horizon

$$p_r = \partial_r \rho \tag{D3}$$

we can now use this result to rewrite the equation for p_ϕ

$$-\frac{\partial_r p_\phi}{r} + \partial_r^2 p_\phi - \partial_r \left(\frac{p_\phi \partial_r \rho}{\rho} \right) - \frac{p_\phi \partial_r \rho}{r\rho} = 0 \tag{D4}$$

using $g^{\phi\phi} p_\phi = p^\phi$

$$-\frac{\partial_r r^2 p^\phi}{r} + \partial_r^2 r^2 p^\phi - \partial_r \left(\frac{r^2 p^\phi \partial_r \rho}{r} \right) - \frac{r p^\phi \partial_r \rho}{\rho} = 0 \tag{D5}$$

which can be rewritten as

$$\partial_r \left(\rho r^3 \partial_r \frac{p^\phi}{\rho} \right) = 0 \tag{D6}$$

integrating two times we obtain

$$\frac{p^\phi}{\rho} = k_1 \int \frac{dr}{\rho r^3} + k_2 \tag{D7}$$

imposing p^ϕ and ρ finite for $r \to \infty$ sets $k_1 = 0$ and $\frac{p^\phi}{\rho} = k_2$, k_2 is the angular velocity of the horizon, then $k_2 \equiv \Omega$ and we can now write

$$p_\phi = \rho r^2 \Omega \tag{D8}$$

Finally we can focus on the equation for p_r which can now be rewritten as follows

$$\begin{aligned} 0 = \partial_r \rho + \partial_r^3 \rho + \frac{\partial_r^2 \rho}{r} - \frac{\partial_r \rho}{r^2} - \partial_r \left(\frac{(\partial_r \rho)^2}{\rho} \right) \\ - \frac{(\partial_r \rho)^2}{\rho r} + \frac{\rho^2 r^4 \Omega^2}{r^3 \rho}. \end{aligned} \tag{D9}$$

applying the change of variable $R(r) = \ln \rho(r)$ we obtain

$$R' + R'R'' + (R')^3 + \frac{R''}{r} - \frac{R'}{r^2} + r\Omega^2 = 0 \tag{D10}$$

integrating once and setting the integration constant to 0 we obtain

$$R^{''} + \frac{R^{'}}{r} + \frac{1}{2}(R^{'})^2 + R + \frac{\Omega^2 r^2}{2} = 0 \tag{D11}$$

The independent term of the equation $\sim \Omega^2 r^2$ is quadratic in r it is then natural to propose an ansatz that is quadratic in r

$$R(r) = A + Br^2 \tag{D12}$$

Substituting in the master equation we find $B = \frac{-1 \pm \sqrt{1-4\Omega^2}}{4}$ and $A = -4B$. Writing $\Omega = \frac{a}{1+a^2}$ and choosing the negative sign in the quadratic equation we arrive at the following solution

$$R(r) = \frac{2}{1+a^2} \left(1 - \frac{r^2}{4} \right) \tag{D13}$$

Appendix E: Explicit derivation of the Black Bar solution

Now let us consider a stationary configuration that is not axisymmetric nor time-independent, and taking a purely rotational velocity field $v^\phi = \Omega$ the equation for $R = R(r, \phi - \Omega t)$ becomes

$$\begin{aligned} 0 = \partial_r^2 R + \frac{\partial_r R}{r} + \frac{\partial_\phi^2 R}{r^2} + \frac{1}{2} \left((\partial_r R)^2 + \frac{(\partial_\phi R)^2}{r^2} \right) \\ + R + \frac{\Omega^2 r^2}{2}. \end{aligned} \tag{E1}$$

which reduces to Eq. (23) for axisymmetric time-independent profiles. The mass density is

$$\rho = e^{R(r, \phi - \Omega t)} \tag{E2}$$

and the momenta are

$$p_r = \partial_r \rho, \qquad p_\phi = \partial_\phi \rho + \Omega r^2 \rho \tag{E3}$$

In order to solve this equation we propose an ansatz of the form $R = R_0 - r^2 F(\phi - \Omega t)$ and put $R_0 = 1$ for convenience. Substituting the ansatz in the equation and

$$\begin{aligned} 0 = \partial_r^2 (r^2 F) + \frac{\partial_r r^2 F}{r} + \partial_\phi^2 F \\ + \frac{1}{2} \left((\partial_r r F)^2 + r^2 (\partial_\phi F)^2 \right) - 1 + r^2 F + \frac{\Omega^2 r^2}{2}. \end{aligned} \tag{E4}$$

at $r = 0$

$$F^{''} + 4F - 1 = 0 \tag{E5}$$

the solution of the homogeneous equation is

$$F_\lambda = A \cos(2(\phi - \Omega t)) + B \sin(2(\phi - \Omega t)) \tag{E6}$$

the general solution will be $F_\lambda + F_p$ where F_p is a particular solution of the equation. Taking F_p to be constant

$$0 + 4F_p = 1 \to F_p = \frac{1}{4} \tag{E7}$$

then

$$F = \frac{1}{4} + A \cos(2(\phi - \Omega t)) + B \sin(2(\phi - \Omega t)) \tag{E8}$$

that we can rewrite as

$$F = \frac{1}{4} + C \cos(2(\phi - \Omega t)) \tag{E9}$$

inserting F back in the equation for $r \neq 0$ we can determine $C = \pm \frac{\sqrt{1-4\Omega^2}}{4}$ so we finally obtain

$$R(t, r, \phi) = 1 - \frac{r^2}{4} \left(1 + \sqrt{1 - 4\Omega^2} \cos(2(\phi - \Omega t)) \right) \tag{E10}$$

and the corresponding mass density

$$\rho(t, r, \phi) = \exp \left[1 - \frac{r^2}{4} (1 + \sqrt{1 - 4\Omega^2} \cos(2(\phi - \Omega t))) \right] \tag{E11}$$

Appendix F: Explicit derivation of the zero modes

Now we want to study small linearized perturbations of the black hole solutions we have found. We start by adding a perturbation to the Myers-Perry solution

$$\varepsilon\delta R(r)e^{i\rho_\phi(\phi-\Omega t)} \tag{F1}$$

where ε is an infinitesimal parameter. When $\rho_\phi = 0$ the perturbations are time-independent and axisymmetric and therefore they are zero modes.
Zero modes are particularly interesting for they lead to new branches of stationary solutions.
When $\rho_\phi \neq 0$ the frequency of the perturbation is

$$\omega = \rho_\phi \Omega \tag{F2}$$

in this case the perturbation corotates with the black hole and therefore we also consider them zero modes.
Inserting the perturbed solution into Eq. (31) we find

$$\begin{aligned} 0 = &\left[-\frac{1}{1+a^2} - \frac{1}{1+a^2} + \frac{r^2}{2(1+a^2)^2} \right.\\ &\left. + \frac{2}{1+a^2} - \frac{r^2}{2(1+a^2)} + \frac{\Omega^2 r^2}{2}\right] \\ &+ \varepsilon e^{i\rho_\phi(\phi-\Omega t)}\left[\delta R''(r) + \frac{1}{r}\delta R'(r) \right.\\ &\left. - \frac{\rho_\phi^2}{r^2}\delta R(r) - \frac{r}{1+a^2}\delta R'(r) + \delta R(r)\right] \\ &+ \mathcal{O}(\varepsilon^2). \end{aligned} \tag{F3}$$

The independent terms cancel except for $\frac{\Omega^2 r^2}{2}$ and we neglect the terms of the order of ε^2. Adjusting δR with an additive constant to get rid of $\frac{\Omega^2 r^2}{2}$ we obtain

$$\delta R^{''} + \left(\frac{1}{r} - \frac{r}{1+a^2}\right)\delta R^{'} + \left(1 - \frac{\rho_\phi^2}{r^2}\right)\delta R = 0 \tag{F4}$$

This equation has a solution in terms of the associated Laguerre polynomials $L_k^{|\rho_\phi|}(x)$

$$\delta R(r) = r^{|\rho_\phi|} L_k^{|\rho_\phi|}\left(\frac{r^2}{2(1+a^2)}\right) \tag{F5}$$

with $k = \frac{a^2+1-|\rho_\phi|}{2}$. We will have the corotating zero modes for

$$a_c^2 = |\rho_\phi| + 2k - 1 \quad , k = 0, 1, 2, ... \tag{F6}$$

Considering $|\rho_\phi| > 0$ we have zero modes for

$$a = \sqrt{1}, \sqrt{2}, \sqrt{3} \rightarrow \Omega = \frac{1}{2}, \frac{\sqrt{2}}{3}, \frac{\sqrt{3}}{4} \tag{F7}$$

In the axisymmetric time-independent case the solutions are in terms of Laguerre Polynomials

$$\delta R(r) = L_k\left(\frac{r^2}{4k}\right) \tag{F8}$$

and we will have zero modes for

$$a_c^2 = 2k - 1 \quad , k = 2, 3 \tag{F9}$$

It should be noted that for $k = 1$ we don't obtain a new solution but the perturbation simply adds angular momentum to our Myers-Perry solution. In this case we have

$$a = \sqrt{3}, \sqrt{5} \rightarrow \Omega = \frac{\sqrt{3}}{4}, \frac{\sqrt{5}}{6} \tag{F10}$$

We can thus see that the first zero mode from which a new branch of solutions appears is obtained for $a = 1 \rightarrow \Omega = \frac{1}{2} \rightarrow \frac{J}{\rho} = 2$. From this point a branch of solutions corresponding to black bars appears and the Myers-Perry branch becomes metastable.
Now let us consider corotating perturbations for black bar solutions.
Starting from the black bar solution in Cartesian coordinates and adding a perturbation

$$\varepsilon\delta R e^{i\rho_\phi(\phi-\Omega t)} \tag{F11}$$

and introducing the perturbed solution into Eq. (31) again we obtain

$$\nabla^2\delta R - \left(\frac{x}{l_\perp^2}\partial_x + \frac{y}{l_\parallel^2}\partial_y\right)\delta R + \delta R = 0 \tag{F12}$$

We will look for factorized solutions of the form $\delta R = f_x(x) f_y(y)$ that will have to satisfy

$$\begin{aligned} \left(\partial_x^2 - \frac{x}{l_\perp^2}\partial_x + 1\right) f_x &= \frac{n_y}{l_\parallel^2} f_x \\ \left(\partial_y^2 - \frac{y}{l_\parallel^2}\partial_y\right) f_y &= -\frac{n_y}{l_\parallel^2} f_y \end{aligned} \tag{F13}$$

These equations have solutions in terms of Hermite polynomials

$$f_x(x) = H_{n_x}\left(\frac{x}{\sqrt{2}l_\perp}\right) \quad f_y(y) = H_{n_y}\left(\frac{y}{\sqrt{2}l_\parallel}\right) \tag{F14}$$

where

$$n_x = l_\perp^2 - \frac{l_\perp^2}{l_\parallel^2} n_y \tag{F15}$$

with n_x, n_y being non-negative integers. This dictates the discrete values Ω for which we have corotating zero modes

$$\Omega = \frac{\sqrt{1-n_x}\sqrt{n_y-1}}{|n_y - n_x|} \tag{F16}$$

For $n_x = 1$ or $n_y = 1$ we have $\Omega = 0$ and we obtain nothing new. For $n_x = 2$, $n_y = 0$ we recover the Myers-Perry solution. The remaining modes all have $n_x = 0$ which leaves us with

$$\Omega = \frac{\sqrt{n_y - 1}}{n_y} \tag{F17}$$

$$l_\parallel^2 = n_y \qquad , l_\perp^2 = \frac{n_y}{n_y - 1} \tag{F18}$$

and

$$\delta R = H_{n_y}\left(\frac{y}{\sqrt{2n_y}}\right) \tag{F19}$$

For $n_y = 2$ we recover a bar mode of the Myers-Perry black hole with $\Omega = \frac{1}{2}$, thus the new zero modes are

$$n_y = 3, 4, 5 \to \Omega = \frac{\sqrt{2}}{3}, \frac{\sqrt{3}}{4}, \frac{\sqrt{4}}{5} \tag{F20}$$

Appendix G: Numerical Methods

1. Shooting method for the dumbbell branch

In order to reproduce the first dumbbell branch we propose the following ansatz

$$R(x, y) = -\frac{x^2}{2l_\perp^2} + R(y), \tag{G1}$$

with

$$R(y) = 1 - \frac{y^2}{2l_\parallel^2} + u(y). \tag{G2}$$

The black bar corresponds to $u(y) = 0$. Therefore, in order to avoid following it with the shooting, we impose $R(0) \neq 1$, which is to say $u(0) \neq 0$.

The perturbation satisfies

$$u'' - \frac{y}{l_\parallel^2}u' + \frac{1}{2}(u')^2 + u = 0. \tag{G3}$$

For the numerical integration we rewrite it as a first-order system,

$$u' = v, \qquad v' = \frac{y}{l_\parallel^2}v - \frac{1}{2}v^2 - u. \tag{G4}$$

The length scales are given by

$$l_\perp^2 = \frac{2}{1 + \sqrt{1 - 4\Omega^2}}, \qquad l_\parallel^2 = \frac{2}{1 - \sqrt{1 - 4\Omega^2}}. \tag{G5}$$

The branch is followed from the zero mode

$$\Omega_N = \frac{\sqrt{N - 1}}{N}, \tag{G6}$$

with $N = 4$ for the first dumbbell branch.

By reflection symmetry, only the region $y \geq 0$ is integrated. The initial conditions are

$$u(0) = u_0, \qquad u'(0) = 0, \tag{G7}$$

where u_0 is the shooting parameter. For each value of Ω, u_0 is chosen so that the solution remains regular for the largest possible integration range. The resulting solution is then used as a predictor for the next value of Ω.

The angular momentum per unit mass is computed as

$$\frac{J}{\rho} = \Omega\left(l_\perp^2 + \langle y^2\rangle\right), \tag{G8}$$

where

$$\langle y^2\rangle = \frac{\int dy\, y^2 e^{R(y)}}{\int dy\, e^{R(y)}}. \tag{G9}$$

2. Collision simulations

The collision simulations solve the leading-order large-D effective equations for the mass density $\rho(t, x, y)$ and the momentum densities $p_x(t, x, y)$, $p_y(t, x, y)$:

$$\partial_t\rho = \nabla^2\rho - \partial_x p_x - \partial_y p_y, \tag{G10}$$

$$\partial_t p_x = \nabla^2 p_x + \partial_x\rho - \partial_x\left(\frac{p_x^2}{\rho}\right) - \partial_y\left(\frac{p_x p_y}{\rho}\right), \tag{G11}$$

$$\partial_t p_y = \nabla^2 p_y + \partial_y\rho - \partial_x\left(\frac{p_x p_y}{\rho}\right) - \partial_y\left(\frac{p_y^2}{\rho}\right). \tag{G12}$$

Spatial derivatives are computed spectrally using fast Fourier transforms, and, the time evolution is performed with a fourth-order Runge-Kutta method.

The initial configuration consist of two boosted Myers-Perry Gaussian blobs. A single blob centered at (x_0, y_0) is described by

$$\rho_{\rm BH} = A\exp\left[-\frac{(x - x_0)^2 + (y - y_0)^2}{2(1 + a^2)}\right], \tag{G13}$$

with initial momenta

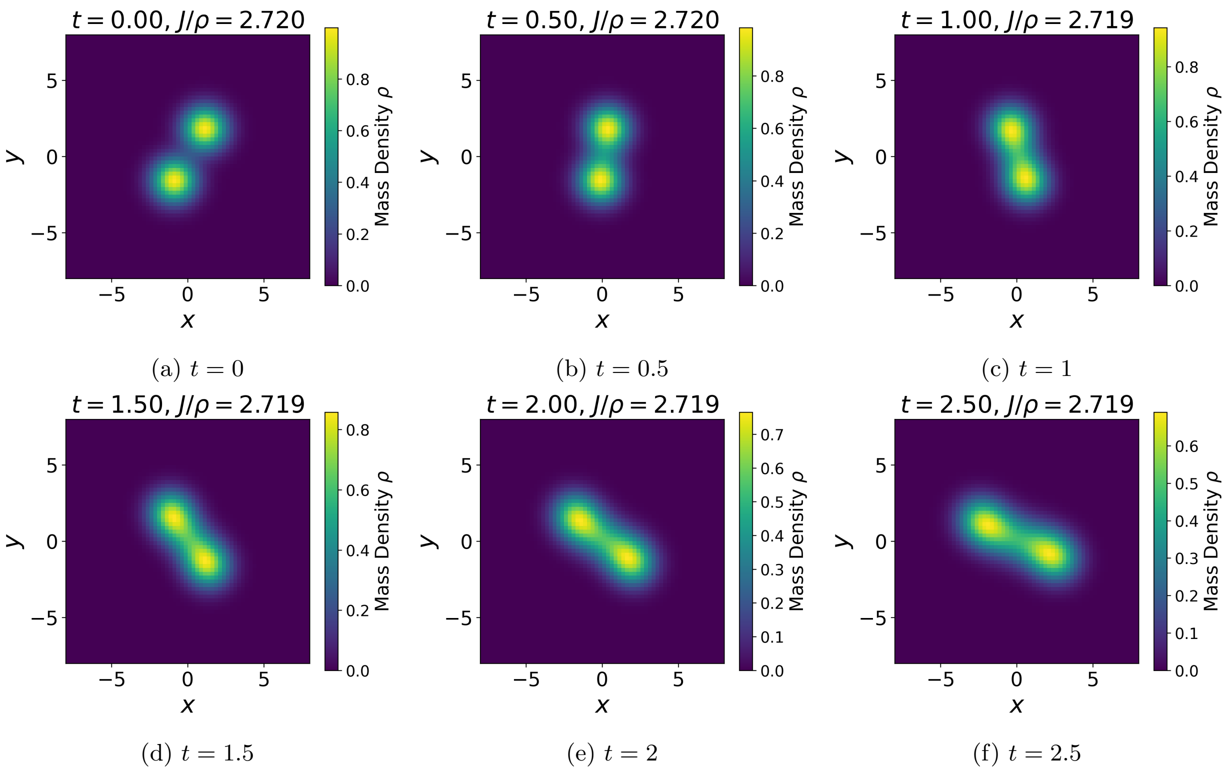
$$p_i = \partial_i\rho_{\rm BH} + \rho_{\rm BH}v_i. \tag{G14}$$

During the evolution we monitor the total mass and angular momentum,

$$\rho_{\rm tot} = \int dx\, dy\, \rho, \qquad J = \int dx\, dy\, (xp_y - yp_x), \tag{G15}$$

and in particular the ratio $J/\rho_{\rm tot}$, denoted in the main text as J/ρ. A small density floor and a velocity cutoff are imposed only as numerical regulators in low-density regions.

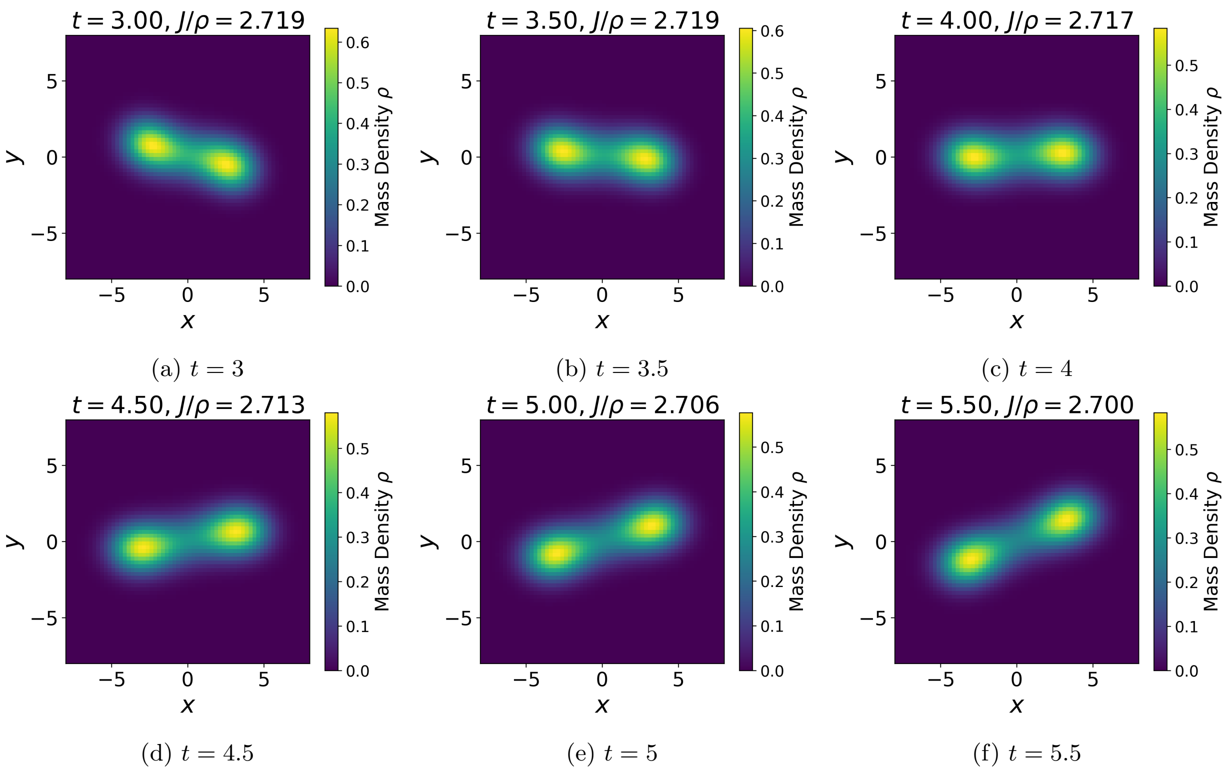
Appendix H: Collision Snapshots



(a) $t = 0$ (b) $t = 0.5$ (c) $t = 1$

(d) $t = 1.5$ (e) $t = 2$ (f) $t = 2.5$

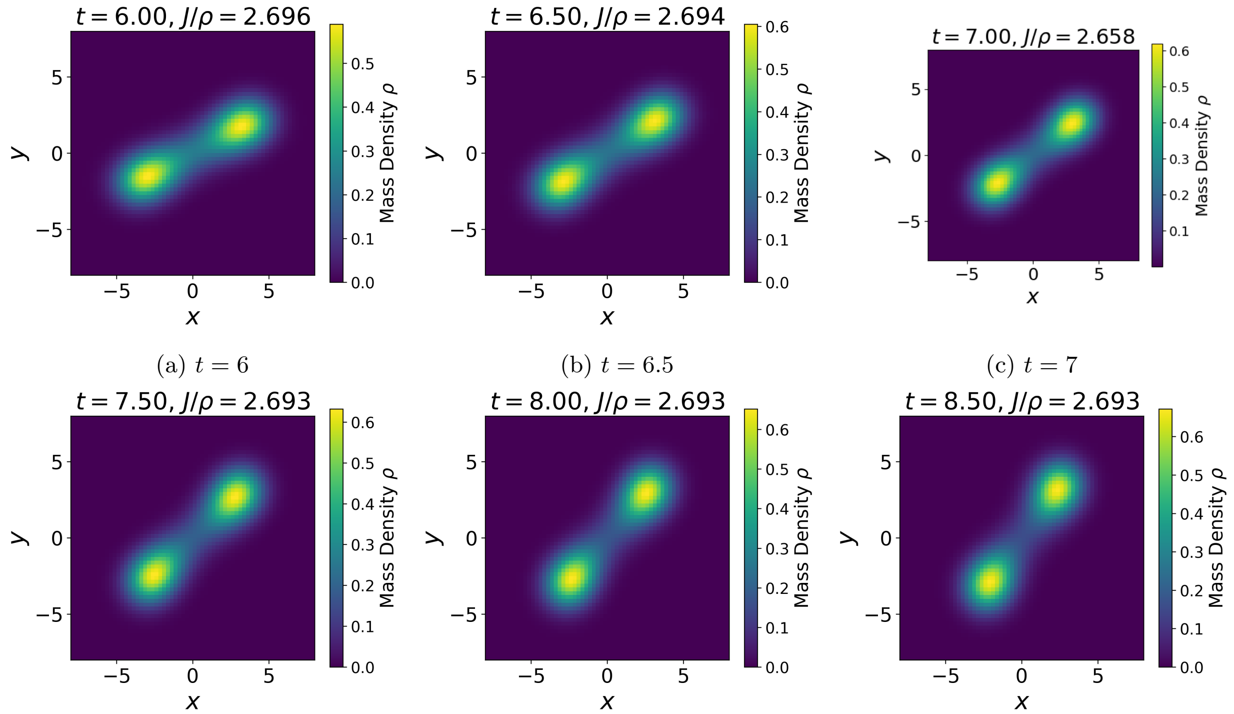
FIG. 4: Evolution of the high-angular-momentum black hole collision for $0 \leq t \leq 2.5$.



(a) $t = 3$ (b) $t = 3.5$ (c) $t = 4$

(d) $t = 4.5$ (e) $t = 5$ (f) $t = 5.5$

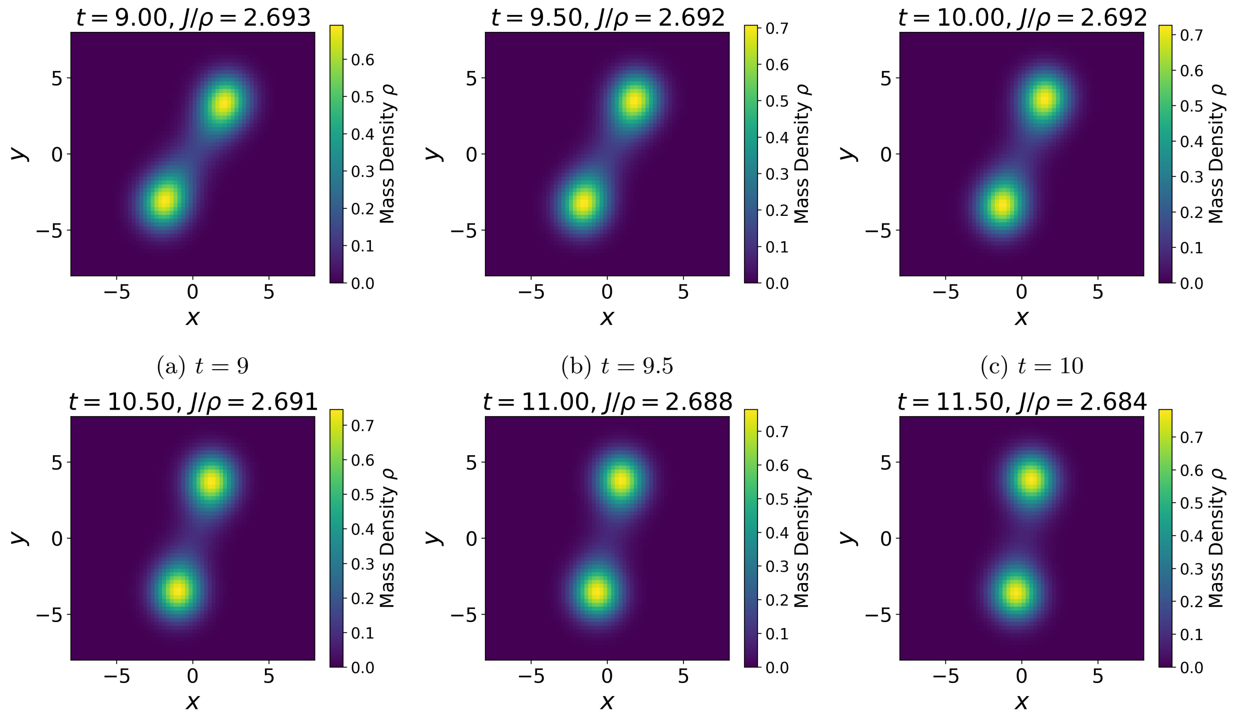
FIG. 5: Evolution of the high-angular-momentum black hole collision for $3 \leq t \leq 5.5$.



(a) $t = 6$ (b) $t = 6.5$ (c) $t = 7$

(d) $t = 7.5$ (e) $t = 8$ (f) $t = 8.5$

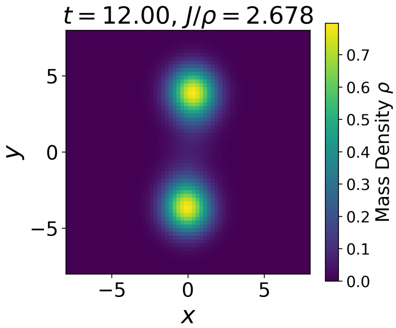
FIG. 6: Evolution of the high-angular-momentum black hole collision for $6 \leq t \leq 8.5$.



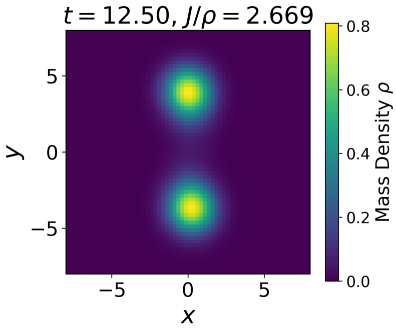
(a) $t = 9$ (b) $t = 9.5$ (c) $t = 10$

(d) $t = 10.5$ (e) $t = 11$ (f) $t = 11.5$

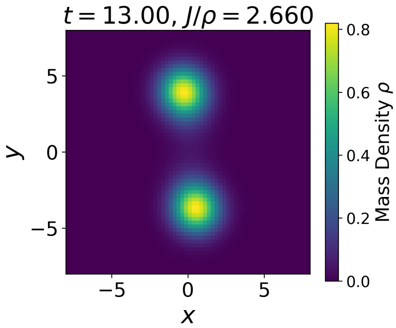
FIG. 7: Evolution of the high-angular-momentum black hole collision for $9 \leq t \leq 11.5$.



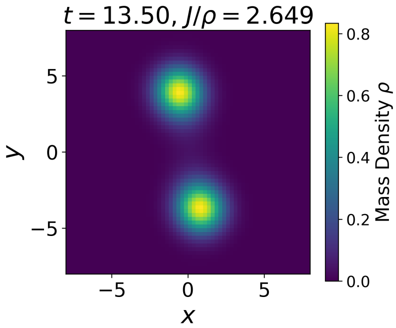
(a) $t = 12$



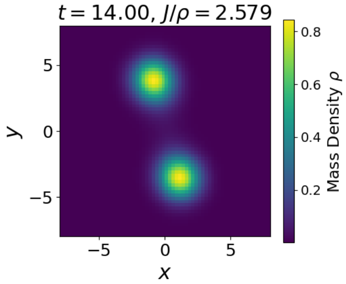
(b) $t = 12.5$



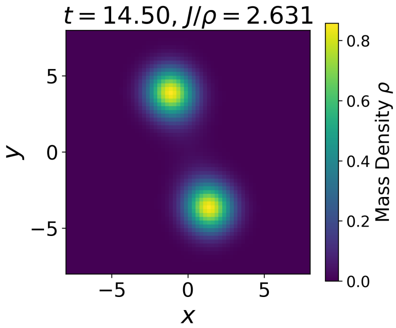
(c) $t = 13$



(d) $t = 13.5$

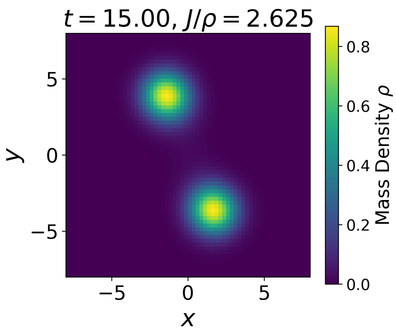


(e) $t = 14$



(f) $t = 14.5$

FIG. 8: Evolution of the high-angular-momentum black hole collision for $12 \leq t \leq 14.5$.



(a) $t = 15$

FIG. 9: Final snapshot of the high-angular-momentum black hole collision.